

Random Forest

1. Introduction

Random Forest is a supervised machine learning algorithm that belongs to the family of ensemble learning methods, specifically Bagging (Bootstrap Aggregation). It constructs multiple decision trees during training and combines their predictions to improve accuracy and reduce overfitting.

2. Definition

Random Forest is an ensemble learning algorithm that builds a collection of decision trees using bootstrap samples and random feature selection, and combines their outputs using majority voting (classification) or averaging (regression).

3. Need for Random Forest

Single decision trees suffer from:

- High variance
- Overfitting
- Sensitivity to small changes in data

Random Forest overcomes these issues by:

- Combining multiple trees
- Introducing randomness in data and features
- Improving generalization performance

4. Ensemble Learning Background

Ensemble learning combines multiple learners to achieve:

- Higher accuracy
- Reduced variance
- Improved robustness

Types of Ensemble Learning:

Method	Description
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Bagging	Parallel training, reduces variance
Boosting	Sequential training, reduces bias
Stacking	Combines heterogeneous models

Random Forest is a Bagging-based ensemble method.

5. Working Principle of Random Forest

Random Forest works by:

1. Creating multiple bootstrap samples from the training dataset.
2. Training a decision tree on each sample.
3. Selecting a random subset of features at each split.
4. Combining predictions of all trees.

6. Random Forest Algorithm:

1. Select N bootstrap samples from the dataset.
2. For each sample:
 - o Grow a decision tree without pruning.
 - o At each node, randomly select m features from total M features.
 - o Choose the best split among the selected features.
3. Repeat until all trees are built.
4. For prediction:
 - o Use majority voting (classification).
 - o Use averaging (regression).

7. Bootstrap Sampling

- Sampling with replacement
- Same instance may appear multiple times
- About 63.2% of data is used in each tree
- Remaining 36.8% is called **Out-of-Bag (OOB) data**

Q. Why ~63.2% of the data is used in each bootstrap sample?

ANS: Assume:

- We have a dataset with **N** observations.
- We create a bootstrap sample by drawing **N times with replacement** from that dataset.

Now let's look at the probability that a **specific observation is NOT selected** in the bootstrap sample.

Step 1: Probability of NOT picking one observation in one draw

Since sampling is uniform:

Step 2: Probability of NOT being picked in N draws

Because we sample **N times independently with replacement**

Step 3: Take the limit as N becomes large

And we know:

So about **36.8%** of the data is **not selected** in a bootstrap sample.

Step 4: Therefore,

So about **63.2%** of the data points appear **at least once** in each bootstrap sample.

CONCLUSION:

In algorithms like Random Forest:

- Each tree trains on ~63.2% unique data points
- The remaining ~36.8% are Out-of-Bag (OOB) samples
- OOB data is used to estimate test error without a separate validation set

8. Random Feature Selection

- At each node, only a subset of features is considered
- Ensures **tree diversity**
- Reduces correlation among trees

Common values:

- Classification: $\sqrt{M}$

- Regression: $M/3$

9. Out-of-Bag (OOB) Error

- OOB samples are not used for training a particular tree
- Used for internal model validation
- Eliminates need for separate validation set

10. Bias-Variance Trade-off

Model	Bias	Variance
Decision Tree	Low	High
Random Forest	Slightly higher	Much lower

Random Forest significantly reduces variance, improving performance.

11. Hyperparameters of Random Forest

Parameter	Description
n_estimators	Number of trees
max_depth	Maximum depth of tree
max_features	Number of features considered
min_samples_split	Minimum samples for split
min_samples_leaf	Minimum samples in leaf

12. Feature Importance

Random Forest provides feature importance by:

- Measuring decrease in impurity
- Permutation importance

Used for:

- Feature selection
- Model interpretation

13. Random Forest for Classification and Regression

Classification:

- Uses **majority voting**
- Output is the class with maximum votes

Regression:

- Uses **average of predictions**
- Output is a continuous value

14. Advantages of Random Forest

- High accuracy
- Handles large datasets
- Reduces overfitting
- Handles missing values
- Works well with high-dimensional data

15. Limitations of Random Forest

- Less interpretable than decision trees
- Computationally expensive
- Requires more memory
- Slower prediction for large forests

16. Applications of Random Forest

- Medical diagnosis
- Fraud detection
- Credit scoring
- Stock market analysis
- Image and text classification

17. Comparison with Other Models

Model	Key Difference
Decision Tree	Single model
Bagging	Uses same features
Random Forest	Uses random features
Boosting	Sequential learning

18. Important Exam Questions

1. Define Random Forest.
2. Explain the working principle of Random Forest.
3. Write the Random Forest algorithm.
4. What is OOB error?
5. Compare Decision Tree and Random Forest.
6. List advantages and limitations of Random Forest.
7. Define Ensemble Learning.
8. Why is Ensemble Learning required?
9. What are the limitations of Decision Trees?
10. Differentiate Bagging and Boosting (basic).
11. Why Random Forest is preferred over Decision Tree?

IMP. Questions

1. What is the bias–variance trade off in machine learning?
2. Explain the bias–variance trade off with reference to Random Forest.
3. How does Random Forest reduce variance compared to a single Decision Tree?

4. Why is Random Forest considered a low-variance ensemble model?
5. What is pruning in Decision Trees?
6. Why is pruning required in Decision Tree learning?
7. Distinguish between pre-pruning and post-pruning.
8. Why are Decision Trees in Random Forest not pruned?
9. How is overfitting controlled in Random Forest without pruning?
10. What is row sampling in ensemble learning?
11. Explain row sampling with replacement and without replacement.
12. What is column sampling in ensemble learning?
13. Explain column sampling with replacement and without replacement.
14. How are row sampling and column sampling applied in Random Forest?
15. Why does Random Forest use row sampling with replacement and column sampling without replacement?
16. What is Out-of-Bag (OOB) data in Random Forest?
17. How is OOB data generated during Random Forest training?
18. Is OOB data wasted? Justify your answer.
19. How is OOB error estimated in Random Forest?
20. Explain how OOB samples can be used for model validation.
21. Compare OOB validation with cross-validation.
22. What are the disadvantages of Random Forest?
23. Why is Random Forest considered less interpretable than Decision Trees?
24. Discuss the computational limitations of Random Forest.
25. In what situations is Random Forest not an ideal choice?
26. Define Out-of-Bag error.
27. State the purpose of bootstrap sampling in Random Forest.
28. State the role of feature randomness in Random Forest.
29. How does ensemble averaging improve model performance?
30. Why does Random Forest generalize better than a single Decision Tree?